JOHNS HOPKINS
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FPGA Design Using VHDL

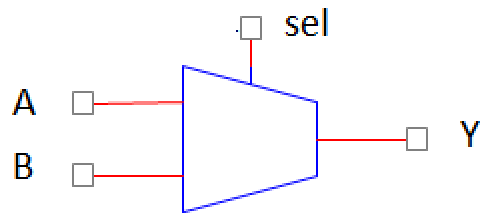
Module 1E: Describing Conditionals in VHDL

Conditional Signal Assignment

How many values can 'sel' take?

- Same syntax as regular assignment with multiple branches:

```
architecture arch of some_entity is
begin
      y <= a when sel = '0' else b;
end arch
```



- **Must end with unconditional else**
- **Priority** implied by syntax is **REAL**
- Another Example:

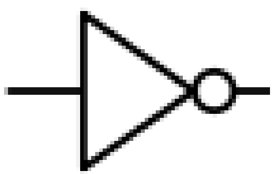
```
conditional_out <= a when sel1 = '0' else
                   b when sel2 = '1' else
                   a xor b when sel3 = '0' else
                   not b;
```

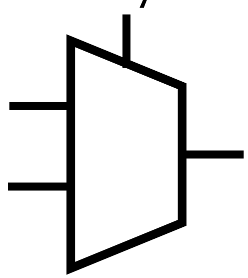
Conditionals

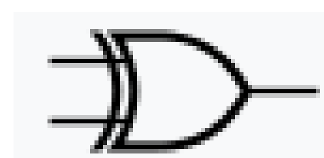
- Possible quiz question:
  - Draw the schematic of the following VHDL code:

```
conditional_out <=
   a when sel1 = '0' else
   b when sel2 = '1' else
   a xor b when sel3 = '0' else
   not b;
```

  - Use the following components in your solution:







Type: std_logic_vector

- **`std_logic_vector`** is simply an array where each element is of type **`std_logic`**

```
entity mux2_8bit is port (
     bus_a : in std_logic_vector(7 downto 0);
     bus_b : in std_logic_vector(7 downto 0);
     sel  : in std_logic;
     bus_out : out std_logic_vector(7 downto 0));
end mux2_8bit;
```

- Individual bits or ranges can be indexed in the architecture:

```
architecture arch of mux2_8bit is
begin
  bus_out(7) <= bus_a(7) when sel = '0' else bus_b(7);
  bus_out(6) <= bus_a(6) when sel = '0' else bus_b(6);
  bus_out(5 downto 0) <= bus_a(5 downto 0) when sel = '0' else
                         bus_b(5 downto 0);
end arch;
```

- Or can be assigned with as an entire vector:

```
bus_out <= bus_a when sel = '0' else bus_b;
```

Selected Signal Assignment

```
entity select_ex is port (
    sel      : in  std_logic_vector(3 downto 0);
    out_disp : out std_logic_vector(3 downto 0))
end select_ex;

architecture arch of select_ex is
begin

-- No priority inferred
-- Need to end with others
with sel select
  out_disp <= "1001" when "0000",
              "0011" when "0001",
              "1101" when "0010",
              "1111" when "0011",
              "1100" when others;
end arch;
```